

EN3160 Assignment 1 on Intensity Transformations and Neighborhood Filtering

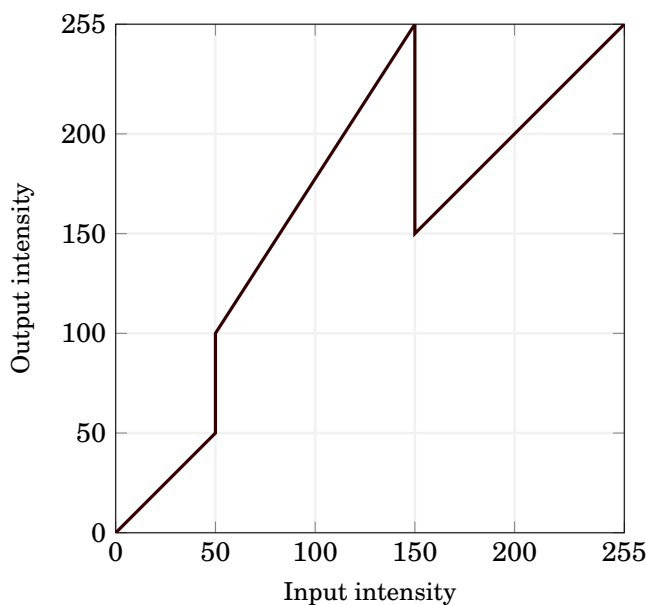
Ranga Rodrigo

September 2, 2026

1. Implement the intensity transformation depicted in Fig. 1a on the image shown in Fig. 1b by writing a function `intensity_transform(im, breakpoints)` where the break points are, e.g.,

$$\begin{bmatrix} 0 & 0 \\ 50 & 50 \\ 100 & 150 \\ 150 & 255 \\ 255 & 255 \end{bmatrix}.$$

Experiment with different break points to get a visually pleasing output. Show the intensity transformation as a plot and the original and transformed images side-by-side.



(a) Intensity transformation.



(b) Image for intensity transformation.

2. Apply a similar operation as above (question 1) to accentuate

- (a) white matter
- (b) gray matter

in the brain proton density image shown in Fig. 2. Show the intensity transformations as a plots.

3. Consider the image shown in Fig. 3¹.

- (a) Apply gamma correction to the L plane in the $L^*a^*b^*$ color space and state the γ value.
- (b) Show the histograms of the original and corrected images.

¹<https://www.adobe.com/creativecloud/photography/discover/highlights-and-shadows.html>

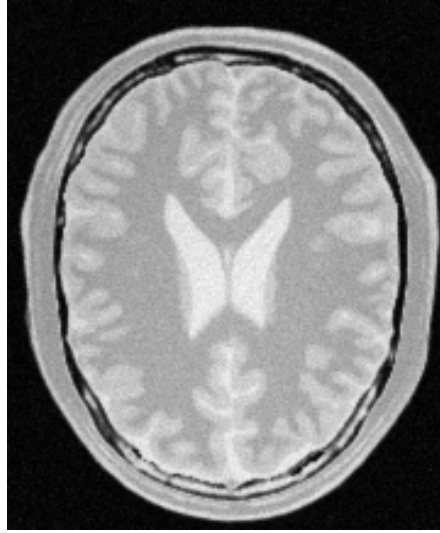


Figure 2: A brain proton density slice.



Figure 3: Image for gamma correction.

4. Increasing the vibrance of a photograph is probably achieved by applying an intensity transformation such as

$$f(x) = \min \left(x + a \times 128 e^{-\frac{(x-128)^2}{2\sigma^2}}, 255 \right),$$

to the saturation plane, where x is the input intensity, $a \in [0, 1]$ and $\sigma = 70$.

- Split the image shown in Fig. 4 into hue, saturation, and value planes.
- Apply the aforementioned intensity transformation to the saturation plane.
- Adjust a to get a visually pleasing output. Report the value of a .
- Recombine the three planes.
- Display the original image, vibrance-enhanced image, and the intensity transformation.



Figure 4: Image for enhancing the vibrance.

- Write a function of your own to carry out histogram equalization on the image shown in Fig. 5. Show the histograms before and after equalization.
- In this question, we will apply histogram equalization only to the foreground of an image to produce an image with a histogram equalized foreground.

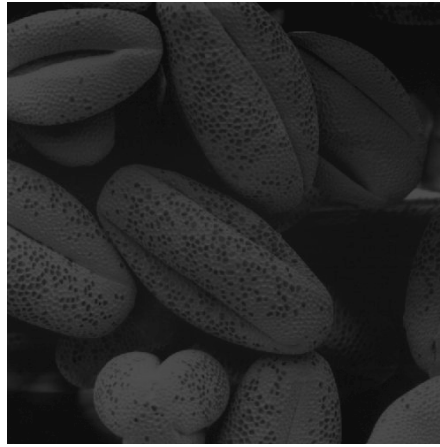


Figure 5: Image for histogram equalization.

- (a) Open the image in Fig. 6, split it into hue, saturation, and values and display these planes in grayscale.
- (b) Select the appropriate plane to threshold in extract the foreground mask. A mask is a binary image.
- (c) Now obtain the foreground only using `cv.bitwise_and` and compute the histogram.
- (d) Obtain the cumulative sum of the histogram using `np.cumsum`.
- (e) Use the formulas in slides to histogram-equalize the foreground.
- (f) Extract the background and add with the histogram equalized foreground.

Show the hue, saturation, and value plane, the mask, the original image, and the result with the histogram-equalized foreground.



Figure 6: Image for histogram equalizing the foreground.

7. Filtering with the Sobel operator can compute the gradient. Consider the image shown in Fig. 7

- (a) Using the existing `filter2D` to Sobel filter the image.
- (b) Write your own code to Sobel filter the image.
- (c) Using the property

$$\begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} * \begin{bmatrix} 1 & 0 & -1 \end{bmatrix},$$

carry out Sobel filtering.

8. Write a program to zoom images by a given factor $s \in (0, 10]$. You must use a function to zoom the image, which can handle

- (a) nearest-neighbor, and
- (b) bilinear interpolation.

I have included four images, two large originals, and there zoomed-out versions. Test your algorithm by computing the normalized sum of squared difference (SSD) when you scale-up the given small images by a factor of 4 by comparing with the original images.

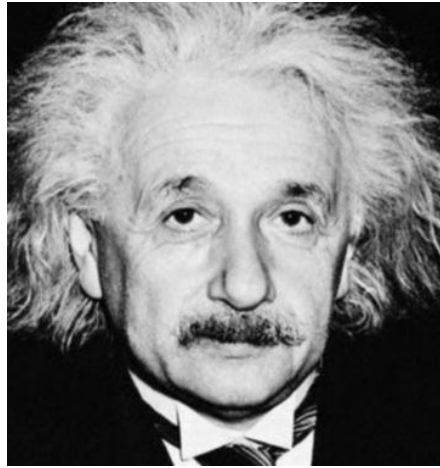


Figure 7: Image for Sobel filtering.

9. Fig. 8² shows a flower image with both the foreground and background are in focus.

- (a) Use grabCut to segment the image. Show the final segmentation mask, foreground image, and background image.
- (b) Produce an enhanced image with a substantially blurred background. Display the original image alongside the enhanced image.
- (c) Why is the background just beyond the edge of the flower quite dark in the enhanced image?



Figure 8: Image enhancing.

10. Fig. 9 shows an image of Lake Minaret. The requirement is to smoothen the image while keeping the edges sharp.

- (a) Use OpenCV's `bilateralFilter` to smoothen the image with a bilateral filter with σ_s and σ_r of your choice. Display the original, Gaussian blurred image with a similar kernel, and bilateral filtered images side-by-side.
- (b) Implement your own bilateral filter and compare the results with OpenCV's `bilateralFilter`. Display the original, Gaussian blurred image with a similar kernel, and bilateral filtered images side-by-side. Use a suitable quantitative measure to compare the results.

²<https://steemit.com/marguerite/ctrl-alt-nwo/marguerite-daisy>



Figure 9: Image bilateral filtering.

GitHub Profile

You must **include the link to your GitHub** (or some other SVN) profile, so that I can see that you have worked on this assignment over a reasonable duration. Therefore, make commits regularly.

Submission

The report must be a **PDF exported directly from your Jupyter Notebook**. Reports created by manually copying and pasting code snippets and images into another document will not be accepted. **Upload a report (maximum 10 pages) named your_index_a01.pdf**. Include **your index number and name within the PDF** as well. The report **should contain the key parts of your code, representative image results, comparisons of results**, and a clear interpretation and **discussion of the findings**. The interpretation of the results and the discussion are important components of the report. An extra-page penalty of 20 marks per page will be applied for pages beyond the 10-page limit.

Grading policy: This assignment contributes **5% to the final grade**. Since **LLMs can be used to generate code**, **you are required to include a link to your GitHub profile in the report**. We will review your commit history and timestamps to verify that the assignment was completed over a reasonable period. Therefore, commit your work regularly throughout the assignment. **For efficiency, only the submitted PDF will be graded**. However, your thorough and detailed understanding of the assignment will be assessed in the final examination through questions about your implementation and results.